# Laporan Eksplorasi Ekosistem Minilab Big Data

*Diajukan untuk penilaian mata kuliah Big Data dan Analitik*


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**PROGRAM STUDI SISTEM INFORMASI**

**FAKULTAS ILMU KOMPUTER**

**UNIVERSITAS BRAWIJAYA**

**MALANG**


**2025**

# TAHAP I
# INGESTION

## 1.1 Environment

Proses deployment dan ingestion data dilakukan secara lokal dengan spesifikasi environment sebagai berikut:

- Sistem Operasi:
  - Microsoft Windows 10 (Version 10.0.26200.8037)
- RAM / Spesifikasi Perangkat:
  - Device name Cassini
  - Processor 12th Gen Intel(R) Core(TM) i5-1235U (1.30 GHz)
  - Installed RAM 16,0 GB (15,7 GB usable)
  - System type 64-bit operating system, x64-based processor
- Tools yang Digunakan:
  - Docker version 29.3.1 (build c2be9cc)
  - Python 3.11
  - PostgreSQL 16 via Docker
  - Object Storage dengan MinIO via Docker
  - Library Python berupa pandas, sqlalchemy, psycopg2-binary, minio, openpyxl, python-dotenv.

## 1.2 Ringkasan Deployment

Deployment tahap awal difokuskan pada penyediaan komponen inti ekosistem data. Dengan menggunakan Docker Compose, dua komponen utama telah berhasil dijalankan:

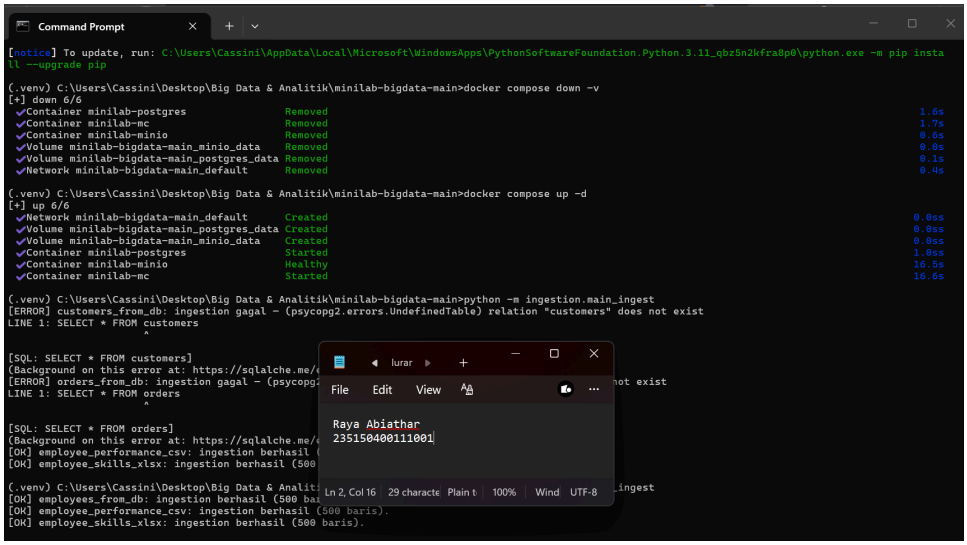
- PostgreSQL (minilab-postgres): Berperan sebagai RDBMS sumber yang menyimpan data transaksional (tabel employee_performance dan employee_skills).
- MinIO (minilab-minio): Berperan sebagai Data Lake (Object Storage) lokal untuk menyimpan data mentah (raw data) hasil ekstraksi, lengkap beserta MinIO Client (minilab-mc) untuk inisialisasi bucket.
- Kedua container berjalan dengan status Up (healthy).

### 1.3 Bukti Hasil Deployment

Berikut adalah bukti bahwa PostgreSQL dan MinIO telah berjalan dengan baik di ekosistem Docker:

- **PostgreSQL**

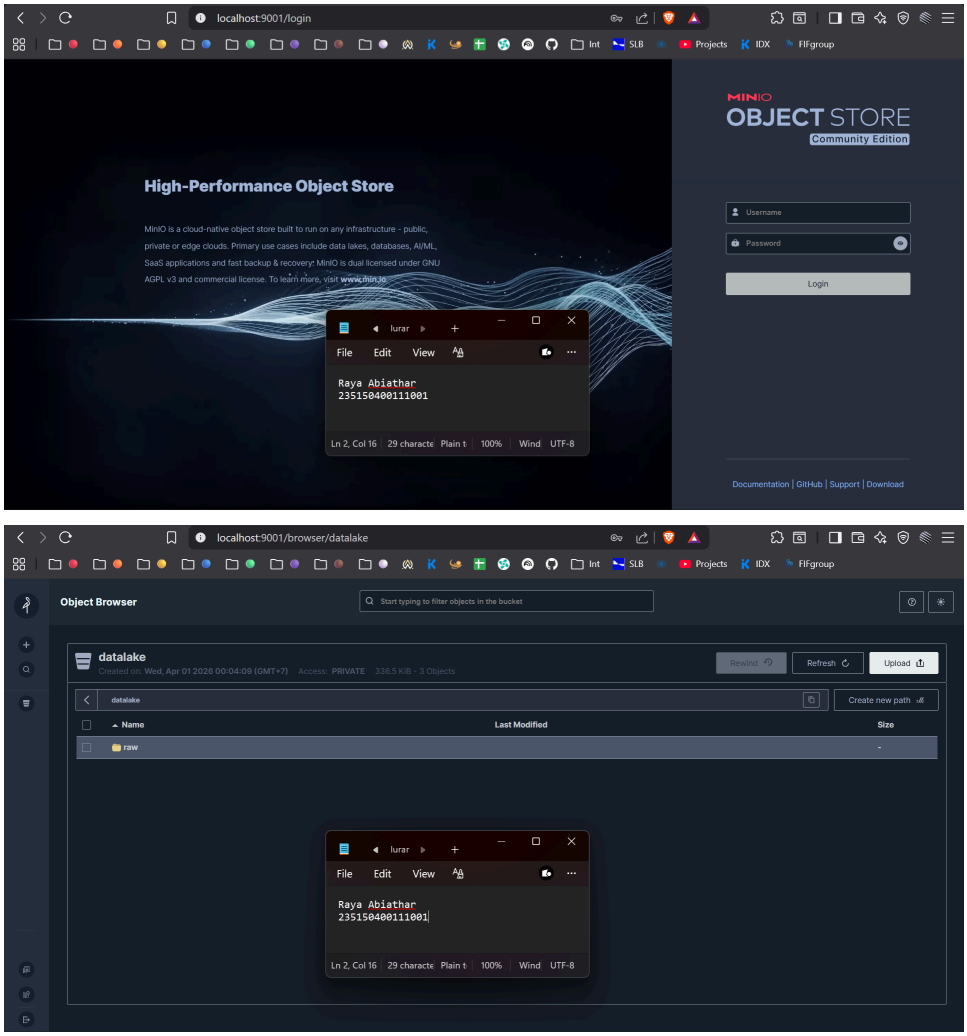
  Berikut adalah dokumentasi untuk penggunaaan PosgreSQL:




**Gambar 1.3.1 Dokumentasi PostgreSQL**

- **MinIO**

  Berikut adalah dokumentasi untuk penggunaaan MinIO:




**Gambar 1.3.2 MinIO**

## 1.4 Bukti Ingestion Data

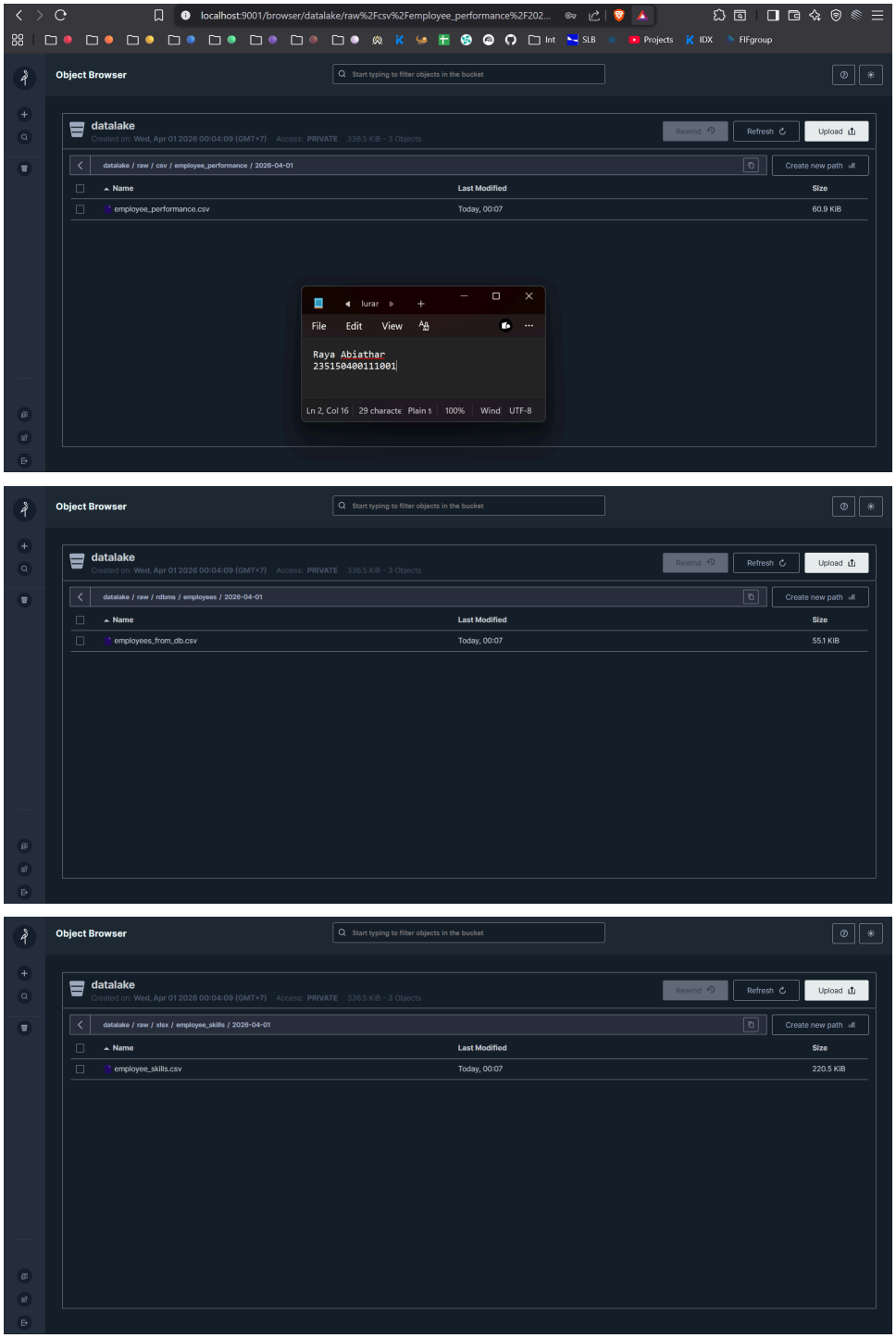
Script ingestion berhasil mengambil data dari RDBMS (PostgreSQL) serta membaca file lokal (CSV dan XLSX), kemudian memuatnya ke dalam MinIO (bucket datalake/raw). Berikut adalah bukti dokumentasi ingestion data:

1. **Log Terminal**

```
(.venv) C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[OK] employees_from_db: ingestion berhasil (500 baris).
[OK] employee_performance_csv: ingestion berhasil (500 baris).
[OK] employee_skills_xlsx: ingestion berhasil (500 baris).
```

**Gambar 1.4.1 Log Terminal Ingestion**

2. **UI MinIO**




**Gambar 1.4.2 Dokumentasi MinIO Ingestion**

## 1.5 Error dan Troubleshooting

Selama proses instalasi dan ingestion, ditemukan dua kendala teknis yang berhasil diatasi:

1. **Kendala DNS/Koneksi saat Pull Image Docker**

| | |
|---|---|
| **Error** | failed to do request: Get "https://production.cloudflare.docker.com/...": dial tcp: lookup production.cloudflare.docker.com: getaddrinfow: A non-recoverable error occurred during a database lookup. |
| **Analysis** | Docker Desktop tidak dapat me-resolve DNS ke registry Docker saat mengunduh image postgres:16 dan minio/mc, kemungkinan akibat cache DNS yang bermasalah pada sistem Windows. |
| **Troubleshooting** | Melakukan verifikasi koneksi internet dengan ping google.com (berhasil), kemudian melakukan flush DNS cache pada Windows dengan perintah ipconfig /flushdns. Setelah itu, proses docker compose up -d kembali diulang dan berhasil menarik image. |

2. **Autentikasi PostgreSQL Gagal (Routing IPv6)**

| | |
|---|---|
| **Error** | [ERROR] customers_from_db: ingestion gagal — (psycopg2.OperationalError) connection to server at "localhost" (::1), port 5432 failed: FATAL: password authentication failed for user "minilab" |
| **Analysis** | Script Python mencoba melakukan koneksi ke RDBMS dengan host localhost. Pada Windows modern, localhost sering kali di-resolve ke alamat loopback IPv6 (::1). Karena port binding Docker biasanya berjalan di IPv4 (127.0.0.1), koneksi ini gagal diteruskan ke dalam container PostgreSQL Docker, atau justru menabrak layanan PostgreSQL lokal bawaan Windows yang berbeda kredensial. |
| **Troubleshooting** | Melakukan penyesuaian konfigurasi host di file sources.yaml. Nilai host: localhost diubah menjadi host: 127.0.0.1 secara eksplisit agar koneksi selalu diarahkan via IPv4. Setelah perubahan tersebut, script berhasil melakukan koneksi dan mengambil data. |

### 1.6 Feedback dan Koreksi

Berdasarkan pengalaman deployment ekosistem ini secara lokal, berikut adalah masukan terhadap pelaksanaan minilab:

- Sangat disarankan untuk mengubah konfigurasi default RDBMS di repository dari localhost menjadi 127.0.0.1.
- Pendekatan menggunakan Docker Compose sangat terstruktur dan cukup ringan. Kombinasi MinIO dan PostgreSQL dapat berjalan stabil di bawah 2GB RAM.

**Lampiran**

Memakai Repository Lama

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Cassini>docker --version
Docker version 29.3.1, build c2be9cc

C:\Users\Cassini>docker compose down -v
no configuration file provided: not found

C:\Users\Cassini>docker compose down -v
no configuration file provided: not found

C:\Users\Cassini>docker compose up -d
no configuration file provided: not found

C:\Users\Cassini>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Cassini>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Cassini>docker compose up -d
no configuration file provided: not found

C:\Users\Cassini>http://localhost:9000
'http:' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Cassini>docker compose up -d
no configuration file provided: not found

C:\Users\Cassini>C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main
'C:\Users\Cassini\Desktop\Big' is not recognized as an internal or external
command,
operable program or batch file.
The system cannot find the path specified.

C:\Users\Cassini>dir
 Volume in drive C is Windows
 Volume Serial Number is 3C69-5647

 Directory of C:\Users\Cassini
```

```
31/03/2026  15:16    <DIR>          .
10/12/2025  11:48    <DIR>          ..
01/01/2026  20:16    <DIR>          .anaconda
31/03/2026  15:16    <DIR>          .cagent
01/01/2026  20:40    <DIR>          .conda
31/03/2026  15:16    <DIR>          .config
01/01/2026  20:32    <DIR>          .continuum
17/02/2026  17:58    <DIR>          .copilot
31/03/2026  22:25    <DIR>          .docker
20/08/2025  09:33                16 .emulator_console_auth_token
15/10/2025  08:17                62 .gitconfig
20/08/2025  09:02    <DIR>          .gradle
31/03/2026  22:19    <DIR>          .ipython
31/03/2026  22:18    <DIR>          .jupyter
25/09/2025  20:45    <DIR>          .skiko
09/04/2025  14:55    <DIR>          .vscode
01/01/2026  20:40    <DIR>          anaconda3
18/11/2025  00:23    <DIR>          AndroidStudioProjects
27/04/2025  03:05    <DIR>          Contacts
09/06/2025  17:49    <DIR>          CrossDevice
31/03/2026  20:25    <DIR>          Desktop
10/03/2026  15:17    <DIR>          Documents
31/03/2026  22:15    <DIR>          Downloads
27/04/2025  03:05    <DIR>          Favorites
31/08/2025  23:28                 0 flutter
31/03/2026  14:05    <DIR>          go
27/04/2025  03:05    <DIR>          Links
27/04/2025  03:05    <DIR>          Music
25/03/2025  23:12    <DIR>          OneDrive
04/10/2025  16:31    <DIR>          Pictures
27/04/2025  03:05    <DIR>          Saved Games
27/04/2025  03:05    <DIR>          Searches
31/08/2025  23:47    <DIR>          source
31/03/2026  15:15    <DIR>          TOS_DI-20211109_1610-V8.0.1
31/03/2026  15:10       801.819.721 TOS_DI-20211109_1610-V8.0.1.zip
10/12/2025  18:11    <DIR>          Videos
               4 File(s)    801.819.799 bytes
              32 Dir(s)  154.748.682.240 bytes free

C:\Users\Cassini>docker compose up -d
no configuration file provided: not found

C:\Users\Cassini>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\Cassini>cd "C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main"

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>dir
```

```
 Volume in drive C is Windows
 Volume Serial Number is 3C69-5647

 Directory of C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main

31/03/2026  22:25    <DIR>          .
31/03/2026  22:25    <DIR>          ..
31/03/2026  22:34               649 .env
31/03/2026  22:24               649 .env.example
31/03/2026  22:24               142 .gitignore
31/03/2026  22:24    <DIR>          analysis
31/03/2026  22:24                21 analysis-requirements.txt
31/03/2026  22:24             2.375 compose.yaml
31/03/2026  22:24    <DIR>          config
31/03/2026  22:24    <DIR>          data
31/03/2026  22:24    <DIR>          ingestion
31/03/2026  22:24    <DIR>          logs
31/03/2026  22:24    <DIR>          notebooks
31/03/2026  22:24           926.234
Penyiapan-Ekosistem-Minilab-Big-Data-untuk-Sumber-Daya-Lokal-Terbatas.pdf
31/03/2026  22:24            10.075 README.md
31/03/2026  22:24               158 requirements.txt
31/03/2026  22:24    <DIR>          tests
               8 File(s)        940.303 bytes
               9 Dir(s)  153.726.013.440 bytes free

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
compose up -d
[+] up 0/3
 - Image minio/minio:latest Pulling
6.0s
 - Image postgres:16        Pulling
6.0s
 - Image minio/mc:latest    Pulling
6.0s
failed to copy: httpReadSeeker: failed open: failed to do request: Get
"https://production.cloudflare.docker.com/registry-v2/docker/registry/v2/bl
obs/sha256/69/69b2ec208575b69597784255eec6fa6a2985ee9e1a47f4411a51f7f5fdd19
3a9/data?expires=1774974389&signature=Xoudmq7yc3F9oDyihtd3nvKgfVA%3D&versio
n=3": dialing production.cloudflare.docker.com:443 container via direct
connection because Docker Desktop has no HTTPS proxy: connecting to
production.cloudflare.docker.com:443: dial tcp: lookup
production.cloudflare.docker.com: getaddrinfow: A non-recoverable error
occurred during a database lookup.

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>ping
google.com

Pinging forcesafesearch.google.com [2001:4860:4802:32::78] with 32 bytes of
data:
Reply from 2001:4860:4802:32::78: time=171ms
Reply from 2001:4860:4802:32::78: time=73ms
Reply from 2001:4860:4802:32::78: time=36ms
Reply from 2001:4860:4802:32::78: time=90ms

Ping statistics for 2001:4860:4802:32::78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 36ms, Maximum = 171ms, Average = 92ms

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>ipconfig
/flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
compose up -d
[+] up 1/19
 - Image minio/mc:latest                Pulling
6.3s
 - Image postgres:16 [⣿                ] Pulling
6.2s
 - Image minio/minio:latest             Pulling
6.2s
failed to copy: httpReadSeeker: failed open: failed to do request: Get
"https://production.cloudflare.docker.com/registry-v2/docker/registry/v2/bl
obs/sha256/69/69b2ec208575b69597784255eec6fa6a2985ee9e1a47f4411a51f7f5fdd19
3a9/data?expires=1774974578&signature=p0FTZ%2F062BwbhtYNogFJ9LBXQiM%3D&vers
ion=3": dialing production.cloudflare.docker.com:443 container via direct
connection because Docker Desktop has no HTTPS proxy: connecting to
production.cloudflare.docker.com:443: dial tcp: lookup
production.cloudflare.docker.com: getaddrinfow: A non-recoverable error
occurred during a database lookup.

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>
C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
pull postgres:16
16: Pulling from library/postgres
4e343c02051d: Pull complete
5b1434d064fd: Pull complete
7c780ba309cf: Pull complete
3db2f2dfc651: Pull complete
```

```
4950166eb819: Pull complete
ec781dee3f47: Pull complete
be939c24ec28: Pull complete
563d41d9b6e2: Pull complete
7f6c666770d4: Pull complete
e060fbc5e5a0: Pull complete
0d650b372b08: Pull complete
eecafd0f9ae3: Pull complete
06d4b5b37c2a: Pull complete
f8e5b8746c3b: Pull complete
2e4d38b98d8b: Download complete
8370d599f21c: Already exists
Digest:
sha256:2586e2a95d1c9b31cb2967feb562948f7d364854453d703039b6efa45fe48417
Status: Downloaded newer image for postgres:16
docker.io/library/postgres:16

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
pull minio/minio:latest
latest: Pulling from minio/minio
34013573f278: Pull complete
1008deaf6ec4: Pull complete
c1bc68842c41: Pull complete
f9c0805c25ee: Pull complete
71e9fc939447: Pull complete
b83ce1c86227: Pull complete
f94d28849fa3: Pull complete
81260b173076: Pull complete
0288b5a0d7e7: Pull complete
Digest:
sha256:14cea493d9a34af32f524e538b8346cf79f3321eff8e708c1e2960462bd8936e
Status: Downloaded newer image for minio/minio:latest
docker.io/minio/minio:latest

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
pull minio/mc:latest
Error response from daemon: failed to resolve reference
"docker.io/minio/mc:latest": failed to do request: Head
"https://registry-1.docker.io/v2/minio/mc/manifests/latest": dialing
registry-1.docker.io:443 container via direct connection because Docker
Desktop has no HTTPS proxy: connecting to registry-1.docker.io:443: dial
tcp: lookup registry-1.docker.io: getaddrinfow: A non-recoverable error
occurred during a database lookup.

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
pull minio/mc:latest
Error response from daemon: failed to resolve reference
"docker.io/minio/mc:latest": failed to do request: Head
"https://registry-1.docker.io/v2/minio/mc/manifests/latest": dialing
```

```
registry-1.docker.io:443 container via direct connection because Docker
Desktop has no HTTPS proxy: connecting to registry-1.docker.io:443: dial
tcp: lookup registry-1.docker.io: getaddrinfow: A non-recoverable error
occurred during a database lookup.

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
pull minio/mc:latest
latest: Pulling from minio/mc
b7c1efd1d2b4: Pull complete
4a93952ab9ce: Pull complete
b268f447c871: Pull complete
f29310fc4a8a: Pull complete
9404e0df6bd3: Pull complete
Digest:
sha256:a7fe349ef4bd8521fb8497f55c6042871b2ae640607cf99d9bede5e9bdf11727
Status: Downloaded newer image for minio/mc:latest
docker.io/minio/mc:latest

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker
compose up -d
[+] up 6/6
 ✔ Network minilab-bigdata-main_default      Created
0.1ss
 ✔ Volume minilab-bigdata-main_postgres_data Created
0.0ss
 ✔ Volume minilab-bigdata-main_minio_data    Created
0.0ss
 ✔ Container minilab-minio                   Healthy
16.8s
 ✔ Container minilab-postgres                Started
1.3ss
 ✔ Container minilab-mc                      Started
16.8s

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED
STATUS                    PORTS
NAMES
6b4695062a7b   minio/mc:latest       "/bin/sh -c ' mc ali…"   19 seconds
ago   Up 1 second
minilab-mc
3996cc1b1c88   postgres:16           "docker-entrypoint.s…"   19 seconds
ago   Up 18 seconds (healthy)   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
minilab-postgres
19880c3729c0   minio/minio:latest    "/usr/bin/docker-ent…"   19 seconds
ago   Up 18 seconds (healthy)   0.0.0.0:9000-9001->9000-9001/tcp,
[::]:9000-9001->9000-9001/tcp   minilab-minio

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>docker ps
```

```
CONTAINER ID   IMAGE                 COMMAND                  CREATED
STATUS                   PORTS
NAMES
6b4695062a7b   minio/mc:latest       "/bin/sh -c ' mc ali…"   2 minutes ago
Up About a minute
minilab-mc
3996cc1b1c88   postgres:16           "docker-entrypoint.s…"   2 minutes ago
Up 2 minutes (healthy)   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
minilab-postgres
19880c3729c0   minio/minio:latest    "/usr/bin/docker-ent…"   2 minutes ago
Up 2 minutes (healthy)   0.0.0.0:9000-9001->9000-9001/tcp,
[::]:9000-9001->9000-9001/tcp   minilab-minio

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>python -m
venv .venv

C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>.venv\Scripts\activatepip install -r
requirements.txt
'.venv\Scripts\activatepip' is not recognized as an internal or external
command,
operable program or batch file.

C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>^A

C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>.venv\Scripts\activate
The system cannot find the path specified.

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>.venv\Scripts\activate
The system cannot find the path specified.

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>(.venv) pip install -r requirements.txt
pip was unexpected at this time.

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>pip install -r requirements.txt
Collecting pandas==2.2.3 (from -r requirements.txt (line 1))
  Using cached pandas-2.2.3-cp311-cp311-win_amd64.whl.metadata (19 kB)
Collecting sqlalchemy==2.0.36 (from -r requirements.txt (line 2))
  Downloading SQLAlchemy-2.0.36-cp311-cp311-win_amd64.whl.metadata (9.9 kB)
Collecting psycopg2-binary==2.9.10 (from -r requirements.txt (line 3))
  Downloading psycopg2_binary-2.9.10-cp311-cp311-win_amd64.whl.metadata
(5.0 kB)
Collecting openpyxl==3.1.5 (from -r requirements.txt (line 4))
  Downloading openpyxl-3.1.5-py2.py3-none-any.whl.metadata (2.5 kB)
Collecting minio==7.2.8 (from -r requirements.txt (line 5))
```

```
  Downloading minio-7.2.8-py3-none-any.whl.metadata (6.5 kB)
Collecting pyyaml==6.0.2 (from -r requirements.txt (line 6))
  Downloading PyYAML-6.0.2-cp311-cp311-win_amd64.whl.metadata (2.1 kB)
Collecting python-dotenv==1.0.1 (from -r requirements.txt (line 7))
  Downloading python_dotenv-1.0.1-py3-none-any.whl.metadata (23 kB)
Collecting pytest==8.3.5 (from -r requirements.txt (line 8))
  Downloading pytest-8.3.5-py3-none-any.whl.metadata (7.6 kB)
Collecting python-dateutil==2.9.0 (from -r requirements.txt (line 9))
  Downloading python_dateutil-2.9.0-py2.py3-none-any.whl.metadata (8.3 kB)
Collecting numpy>=1.23.2 (from pandas==2.2.3->-r requirements.txt (line 1))
  Downloading numpy-2.4.4-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Collecting pytz>=2020.1 (from pandas==2.2.3->-r requirements.txt (line 1))
  Downloading pytz-2026.1.post1-py2.py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: tzdata>=2022.7 in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
pandas==2.2.3->-r requirements.txt (line 1)) (2025.3)
Requirement already satisfied: typing-extensions>=4.6.0 in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
sqlalchemy==2.0.36->-r requirements.txt (line 2)) (4.15.0)
Collecting greenlet!=0.4.17 (from sqlalchemy==2.0.36->-r requirements.txt
(line 2))
  Downloading greenlet-3.3.2-cp311-cp311-win_amd64.whl.metadata (3.8 kB)
Collecting et-xmlfile (from openpyxl==3.1.5->-r requirements.txt (line 4))
  Downloading et_xmlfile-2.0.0-py3-none-any.whl.metadata (2.7 kB)
Requirement already satisfied: certifi in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
minio==7.2.8->-r requirements.txt (line 5)) (2026.2.25)
Requirement already satisfied: urllib3 in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
minio==7.2.8->-r requirements.txt (line 5)) (2.6.3)
Requirement already satisfied: argon2-cffi in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
minio==7.2.8->-r requirements.txt (line 5)) (25.1.0)
Collecting pycryptodome (from minio==7.2.8->-r requirements.txt (line 5))
  Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl.metadata (3.5 kB)
Requirement already satisfied: colorama in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
pytest==8.3.5->-r requirements.txt (line 8)) (0.4.6)
Collecting iniconfig (from pytest==8.3.5->-r requirements.txt (line 8))
  Downloading iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Requirement already satisfied: packaging in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
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pytest==8.3.5->-r requirements.txt (line 8)) (26.0)
Collecting pluggy<2,>=1.5 (from pytest==8.3.5->-r requirements.txt (line
8))
  Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: six>=1.5 in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
python-dateutil==2.9.0->-r requirements.txt (line 9)) (1.17.0)
Requirement already satisfied: argon2-cffi-bindings in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
argon2-cffi->minio==7.2.8->-r requirements.txt (line 5)) (25.1.0)
Requirement already satisfied: cffi>=1.0.1 in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
argon2-cffi-bindings->argon2-cffi->minio==7.2.8->-r requirements.txt (line
5)) (2.0.0)
Requirement already satisfied: pycparser in
c:\users\cassini\appdata\local\packages\pythonsoftwarefoundation.python.3.1
1_qbz5n2kfra8p0\localcache\local-packages\python311\site-packages (from
cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi->minio==7.2.8->-r
requirements.txt (line 5)) (3.0)
Using cached pandas-2.2.3-cp311-cp311-win_amd64.whl (11.6 MB)
Downloading SQLAlchemy-2.0.36-cp311-cp311-win_amd64.whl (2.1 MB)

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2.1/2.1 MB 7.3 MB/s eta 0:00:00
Downloading psycopg2_binary-2.9.10-cp311-cp311-win_amd64.whl (1.2 MB)

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1.2/1.2 MB 6.4 MB/s eta 0:00:00
Downloading openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)
Downloading minio-7.2.8-py3-none-any.whl (93 kB)
Downloading PyYAML-6.0.2-cp311-cp311-win_amd64.whl (161 kB)
Downloading python_dotenv-1.0.1-py3-none-any.whl (19 kB)
Downloading pytest-8.3.5-py3-none-any.whl (343 kB)
Downloading python_dateutil-2.9.0-py2.py3-none-any.whl (230 kB)
Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
Downloading greenlet-3.3.2-cp311-cp311-win_amd64.whl (230 kB)
Downloading numpy-2.4.4-cp311-cp311-win_amd64.whl (12.6 MB)

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12.6/12.6 MB 6.5 MB/s eta 0:00:00
Downloading pytz-2026.1.post1-py2.py3-none-any.whl (510 kB)
Downloading et_xmlfile-2.0.0-py3-none-any.whl (18 kB)
Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Downloading pycryptodome-3.23.0-cp37-abi3-win_amd64.whl (1.8 MB)

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```

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1.8/1.8 MB 6.6 MB/s eta 0:00:00
Installing collected packages: pytz, pyyaml, python-dotenv,
python-dateutil, pycryptodome, psycopg2-binary, pluggy, numpy, iniconfig,
greenlet, et-xmlfile, sqlalchemy, pytest, pandas, openpyxl, minio
  Attempting uninstall: pyyaml
    Found existing installation: PyYAML 6.0.3
    Uninstalling PyYAML-6.0.3:
      Successfully uninstalled PyYAML-6.0.3
  Attempting uninstall: python-dateutil
    Found existing installation: python-dateutil 2.9.0.post0
    Uninstalling python-dateutil-2.9.0.post0:
      Successfully uninstalled python-dateutil-2.9.0.post0
Successfully installed et-xmlfile-2.0.0 greenlet-3.3.2 iniconfig-2.3.0
minio-7.2.8 numpy-2.4.4 openpyxl-3.1.5 pandas-2.2.3 pluggy-1.6.0
psycopg2-binary-2.9.10 pycryptodome-3.23.0 pytest-8.3.5
python-dateutil-2.9.0 python-dotenv-1.0.1 pytz-2026.1.post1 pyyaml-6.0.2
sqlalchemy-2.0.36

[notice] A new release of pip is available: 25.1.1 -> 26.0.1
[notice] To update, run:
C:\Users\Cassini\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundati
on.Python.3.11_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[ERROR] customers_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[ERROR] orders_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[OK] customers_from_csv: ingestion berhasil (25 baris).
[OK] products_from_xlsx: ingestion berhasil (15 baris).

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose down -v
[+] down 6/6
 ✔ Container minilab-mc                      Removed
1.3s
 ✔ Container minilab-postgres                Removed
0.4s
 ✔ Container minilab-minio                   Removed
0.4s
 ✔ Volume minilab-bigdata-main_postgres_data Removed
0.0s
```

```
 ✔ Volume minilab-bigdata-main_minio_data    Removed
0.0s
 ✔ Network minilab-bigdata-main_default      Removed
0.3s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose up -d
[+] up 6/6
 ✔ Network minilab-bigdata-main_default      Created
0.0ss
 ✔ Volume minilab-bigdata-main_postgres_data Created
0.0ss
 ✔ Volume minilab-bigdata-main_minio_data    Created
0.0ss
 ✔ Container minilab-minio                   Healthy
16.2s
 ✔ Container minilab-postgres                Started
0.7ss
 ✔ Container minilab-mc                      Started
16.3s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[ERROR] customers_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[ERROR] orders_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[OK] customers_from_csv: ingestion berhasil (25 baris).
[OK] products_from_xlsx: ingestion berhasil (15 baris).

docker compose down -v
[+] down 6/6sers\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>
 ✔ Container minilab-postgres                Removed
0.4s
 ✔ Container minilab-mc                      Removed
1.3s
 ✔ Container minilab-minio                   Removed
0.5s
 ✔ Volume minilab-bigdata-main_minio_data    Removed
0.0s
 ✔ Volume minilab-bigdata-main_postgres_data Removed
0.0s
 ✔ Network minilab-bigdata-main_default      Removed
```

```
0.3s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose up -d
[+] up 6/6
 ✔ Network minilab-bigdata-main_default      Created
0.1ss
 ✔ Volume minilab-bigdata-main_postgres_data Created
0.0ss
 ✔ Volume minilab-bigdata-main_minio_data    Created
0.0ss
 ✔ Container minilab-minio                   Healthy
15.8s
 ✔ Container minilab-postgres                Started
0.8ss
 ✔ Container minilab-mc                      Started
15.8s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker exec -it minilab-postgres psql -U
minilab -d minilabdb
psql (16.13 (Debian 16.13-1.pgdg13+1))
Type "help" for help.

minilabdb=# python -m ingestion.main_ingest
minilabdb-# ^\
(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[ERROR] customers_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[ERROR] orders_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[OK] customers_from_csv: ingestion berhasil (25 baris).
[OK] products_from_xlsx: ingestion berhasil (15 baris).

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose down -v
[+] down 6/6
 ✔ Container minilab-mc                      Removed
1.3s
 ✔ Container minilab-postgres                Removed
0.4s
 ✔ Container minilab-minio                   Removed
```

```
0.5s
 ✔ Volume minilab-bigdata-main_postgres_data Removed
0.0s
 ✔ Volume minilab-bigdata-main_minio_data    Removed
0.0s
 ✔ Network minilab-bigdata-main_default      Removed
0.2s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose up -d
[+] up 6/6
 ✔ Network minilab-bigdata-main_default      Created
0.1ss
 ✔ Volume minilab-bigdata-main_postgres_data Created
0.0ss
 ✔ Volume minilab-bigdata-main_minio_data    Created
0.0ss
 ✔ Container minilab-postgres                Started
0.7ss
 ✔ Container minilab-minio                   Healthy
15.8s
 ✔ Container minilab-mc                      Started
15.9s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker exec -it minilab-postgres psql -U
minilab -d minilabdb
psql (16.13 (Debian 16.13-1.pgdg13+1))
Type "help" for help.

minilabdb=# python -m ingestion.main_ingest
(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[ERROR] customers_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[ERROR] orders_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[OK] customers_from_csv: ingestion berhasil (25 baris).
[OK] products_from_xlsx: ingestion berhasil (15 baris).

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingesttimeout /t 30
C:\Users\Cassini\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundati
```

```
on.Python.3.11_qbz5n2kfra8p0\python.exe: No module named
ingestion.main_ingesttimeout

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>timeout /t 30

Waiting for  0 seconds, press a key to continue ...

(.venv) C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>
(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[ERROR] customers_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[ERROR] orders_from_db: ingestion gagal — (psycopg2.OperationalError)
connection to server at "localhost" (::1), port 5432 failed: FATAL:
password authentication failed for user "minilab"

(Background on this error at: https://sqlalche.me/e/20/e3q8)
[OK] customers_from_csv: ingestion berhasil (25 baris).
[OK] products_from_xlsx: ingestion berhasil (15 baris).

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[OK] customers_from_db: ingestion berhasil (30 baris).
[OK] orders_from_db: ingestion berhasil (63 baris).
[OK] customers_from_csv: ingestion berhasil (25 baris).
[OK] products_from_xlsx: ingestion berhasil (15 baris).

(.venv) C:\Users\Cassini\Desktop\Big Data & Analitik\minilab-bigdata-main>
```

Memakai Dataset Baru (Employee Burnout via Kaggle)

```
(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose down -v
[+] down 6/6
 ✔ Container minilab-postgres                Removed
1.6s
 ✔ Container minilab-mc                      Removed
1.7s
 ✔ Container minilab-minio                   Removed
0.6s
 ✔ Volume minilab-bigdata-main_minio_data    Removed
0.0s
 ✔ Volume minilab-bigdata-main_postgres_data Removed
```

```
0.1s
 ✔ Network minilab-bigdata-main_default      Removed
0.4s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>docker compose up -d
[+] up 6/6
 ✔ Network minilab-bigdata-main_default      Created
0.0ss
 ✔ Volume minilab-bigdata-main_postgres_data Created
0.0ss
 ✔ Volume minilab-bigdata-main_minio_data    Created
0.0ss
 ✔ Container minilab-postgres                Started
1.0ss
 ✔ Container minilab-minio                   Healthy
16.5s
 ✔ Container minilab-mc                      Started
16.6s

(.venv) C:\Users\Cassini\Desktop\Big Data &
Analitik\minilab-bigdata-main>python -m ingestion.main_ingest
[ERROR] customers_from_db: ingestion gagal —
(psycopg2.errors.UndefinedTable) relation "customers" does not exist
LINE 1: SELECT * FROM customers
                      ^

[SQL: SELECT * FROM customers]
(Background on this error at: https://sqlalche.me/e/20/f405)
[ERROR] orders_from_db: ingestion gagal — (psycopg2.errors.UndefinedTable)
relation "orders" does not exist
LINE 1: SELECT * FROM orders
                      ^

[SQL: SELECT * FROM orders]
(Background on this error at: https://sqlalche.me/e/20/f405)
[OK] employee_performance_csv: ingestion berhasil (500 baris).
[OK] employee_skills_xlsx: ingestion berhasil (500 baris).
```